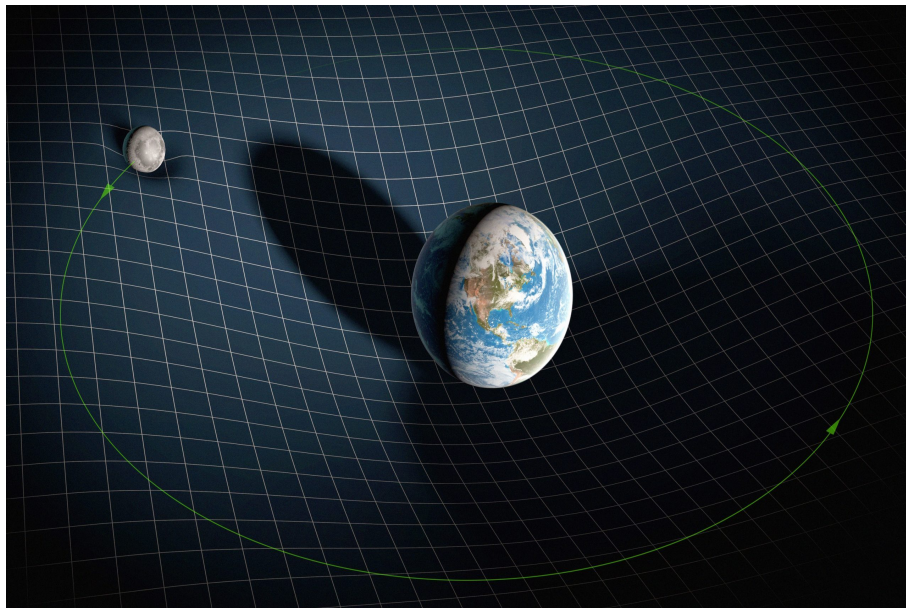




Faculty Of Sciences

Dynamical analysis of a dark-matter halo and galaxies in cosmological, hydrodynamical simulations.



Boateng Christopher, Verheecke Jitske

Master of Science in Physics and Astronomy

Department of Physics and Astronomy - WE05

October 2025

Contents

1	Dynamical analysis of a dark-matter halo in cosmological and hydrodynamical simulations	1
A	Appendix	6

Chapter 1

Dynamical analysis of a dark-matter halo in cosmological and hydrodynamical simulations

Simulation forms an important tool in theoretical astrophysics, allowing to model the complex structure of processes of the universe. Moreover, with the increase in computational power, simulations have become more detailed and accessible for application in astrophysics.

The IllustrisTNG project steers towards a predictive theory of galaxy formation, and aims to model the universe from the very beginning after the Big Bang to the present day. In this project a short and quantitative analysis of a cosmological simulation that was performed within the framework of IllustrisTNG is presented. Here, the focus is laid on the analysis and discussion of the properties of a single subhalo, thus a cutout with ID 344137.

This dataset contains $N = 268525$ dark matter particles with in a cubic volume of side length $L = 35000h^{-1}\text{kpc}$ in physical length at redshift of $z \approx 0$. Here, the quantity $h = 0.6774$ refers to the dimensionless Hubble parameter. Each individual particle the ensemble has a mass of $m_{\text{DM}} \approx 10^{6.73}M_{\odot}$ with $M_{\odot}$ the solar mass, 3D position and corresponding velocity attributes. The system was simulated in a ΛCDM setting with $\Omega_m = 0.3089$ and $\Omega_{\Lambda} = 0.6911$. Even though baryonic processes will not be discussed in this work, it is worth mentioning that the full simulation also included gas cells (baryonic matter) as well with a baryonic content of $\Omega_b = 0.0486$. Other cosmological parameters and quantities of the simulation that are relevant for this work is summarized in table A.1.

Spatial Distribution of DM particles

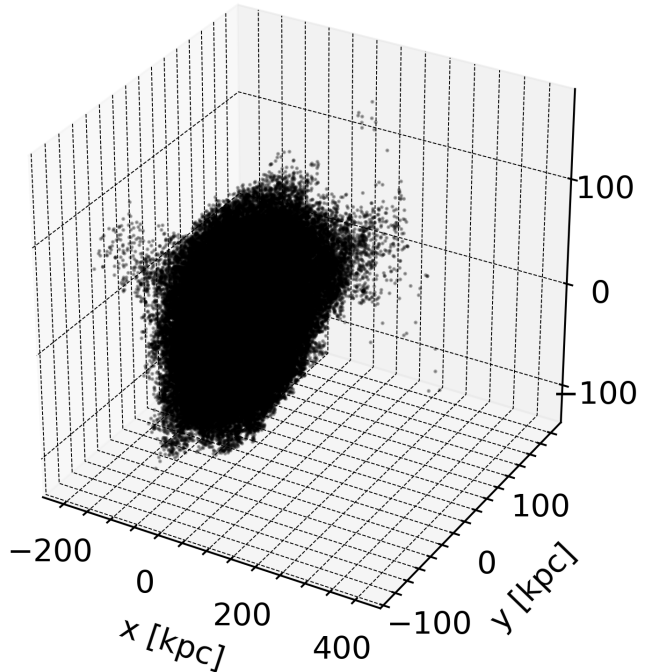


Figure 1.1: 3D distribution of the dark matter particles in the subhalo. The majority of the particles are concentrated within a radius of $r \approx 100$ kpc around the center of mass of the system with few outliers reaching distances up to $r \approx 407$ kpc.

A 3D representation of the dark matter distribution of the cutout in the subhalo, centered around the center of mass of the system, is shown in fig. 1.1. This exposes the extent of the halo

as well as its rough distribution. On scale lengths up to approximately 100 kpc, the data can be considered spherically symmetric, clustered around the center of mass, with a handful of outliers reaching distances up to $r \approx 400$ kpc. The spherical nature of the halo can further be sensed by analysis of the 2D distribution of the particles. As illustrated in fig. 1.2, where the projection along different planes are shown, with color coding of the speed of the particles, the distribution appears roughly uniform in all directions hinting at a spherical system.

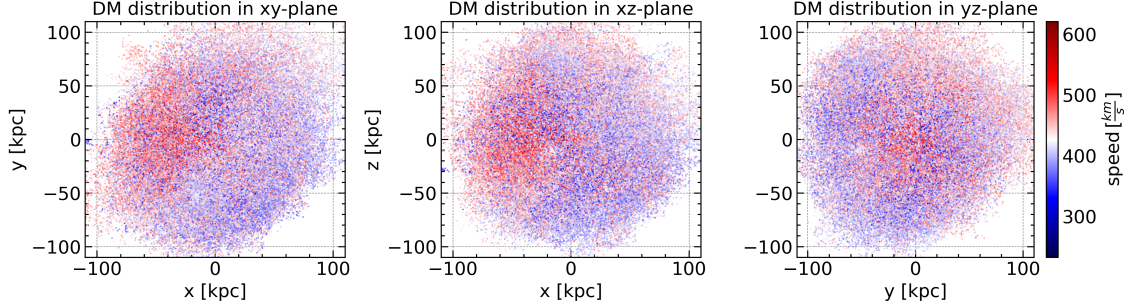


Figure 1.2: Dark matter distribution of the cutout with ID 344137 from the IllustrisTNG simulation at redshift $z \approx 0$ in different projections. The color coding indicates the speed of the particles ranging from 230 km/s to 620 km/s.

This symmetry behaviour can be quantified by decomposition of the distribution of the dataset on the spherical basis as is discussed as follows. The number density profile ($n(\mathbf{r})$) of a distribution of point particles with coordinates $\mathbf{r} = (x, y, z)$ is generally given by

$$n(\mathbf{r}) = n(r, \theta, \phi) = \frac{1}{V} \sum_i \delta(\mathbf{r} - \mathbf{r}_i)$$

where the sum runs over all particles i at position $\mathbf{r}_i$ within a volume V . In the framework, the dataset is binned in $N = 28$ spherical shells of thickness Δr and radius r around the center of mass of the system, containing particles i with spherical coordinates (r_i, θ_i, ϕ_i) . For each shell, the spherical harmonics decomposition of the distribution field is performed by the virtue of chapter 1

$$n(\mathbf{r}) = n(r, \theta, \phi) = \sum_{l=0}^{\infty} \sum_{m=-l}^l n_{lm}(r) Y_{lm}(\theta, \phi)$$

with $Y_{lm}(\theta, \phi)$ the spherical harmonics basis functions of order l and degree m , and invoking the orthonormality. The coefficients $n_{lm}(r)$ are then further used to compute the power spectrum P_l of the distribution as a function of the multipole order l via

$$P_l = \frac{1}{2l+1} \sum_{m=-l}^l |n_{lm}|^2. \quad (1.1)$$

For a perfectly spherically symmetric system, only the monopole term (n_{00}) with $(l = 0, m = 0)$ in the expansion survives. However, as shown in fig. 1.3, the power originating from the dipole ($l = 1$) and quadrupole ($l = 2$) as well as $l = 3$ survives. Eventhough these power survives, it can be seen that this remains in a good approximation constant for distances below approximately 100 kpc and relatively smaller than P_0 . The increase in the power for radii larger than 100 kpc can be attributed to outliers and uncertainties in the density field due to less data points.

Thus, in a good approximation, the dark matter distribution in the subhalo can be considered spherically symmetric, which allows for simplifications in the further analysis of the system. Carrying on with a spherical distribution in mind, the mass density profile ($\rho(r)$), hence forth referred simply to as density, is directly related to the spherical number density via $\rho(r) = m_{\text{DM}} n(r)$ with m_{DM} the mass of an individual dark matter particle. The density $\rho(r)$ or equivalently $n(r)$ of the discrete dataset is obtained by binning the particles in radial distance of radius r and thickness Δr in the center of mass of the system and counting the number of particles in each bin. This is simply achieved by use of the `numpy histogram` function with 28 bins logarithmically spaced between the minimum and maximum radial distance of the particles. Dividing the number of particles in each

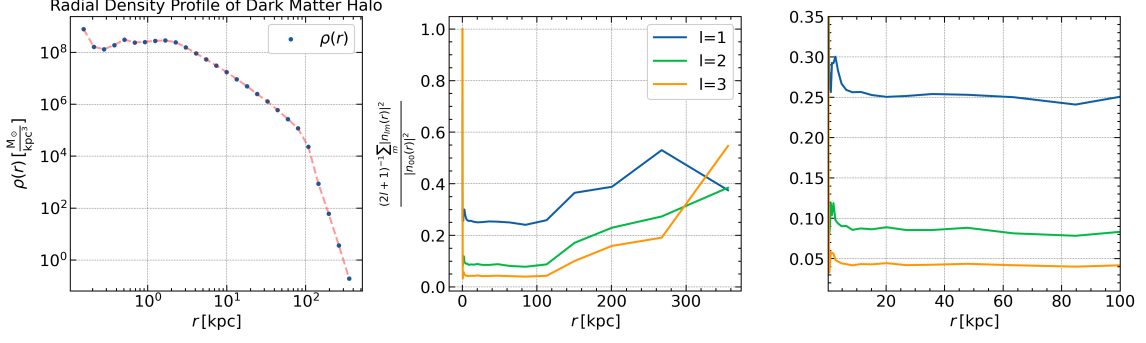


Figure 1.3: Spherical harmonic decomposition of the density field of the dark matter distribution in the subhalo. The power spectrum P_l is normalized to the monopole term P_0 . The dominance of the monopole term ($l = 0$) indicates a high degree of spherical symmetry of the system.

bin by the corresponding shell volume and multiplying with the particle mass yields the density profile $\rho(r)$. Mathematically, this process amount to

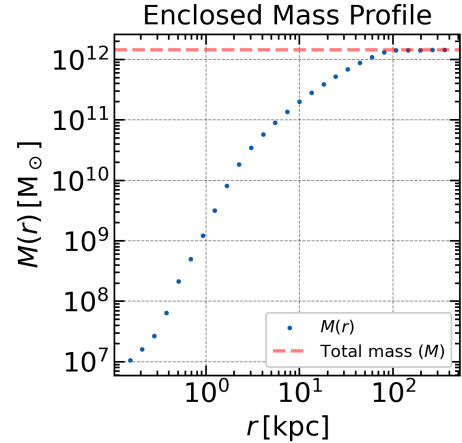
$$\rho(r_i) = \frac{m_{\text{DM}}}{V_i} \sum_{r_j \in \text{shell } i} 1 = \frac{3 \cdot m_{\text{DM}}}{4\pi(r_{i+1}^3 - r_{i-1}^3)} \sum_{r_j \in \text{shell } i} 1. \quad (1.2)$$

The resulting density profile is shown in fig. 1.3 (**left**) on a log-log representation. The density profile shows a typical behaviour of a dark matter halo, with a steep decrease in density for larger radii ($r \geq 100$ kpc) and a cusp, i.e flattening for radii approaching zero. This system has a density of order of $\rho \approx 10^8 M_\odot \text{kpc}^{-3}$ at $r \leq 1$ kpc, decreasing to $\rho \approx 10^{-0.7} M_\odot \text{kpc}^{-3}$ at $r \approx 400$ kpc, encompassing a total mass of $M = 1.45 \times 10^{12} M_\odot$.

$$M(r_i) = 4\pi \sum_{r_j \leq r_i} r_j'^2 \rho(r_j) \Delta r_j \quad (1.3)$$

The enclosed mass profile $M(r)$ of the system, computed via integration of the density profile according to eq. (1.3), is illustrated in fig. 1.4. It is evident as argued already that almost all of the mass of the system is concentrated within a radius of approximately 100 kpc, with virtually no to little increase in mass for larger radii.

Figure 1.4: Enclosed mass profile $M(r)$ of the dark matter distribution in the subhalo as a function of radial distance r . The enclosed mass is computed via numerical integration of the density profile $\rho(r)$.



Continue from here

$$\Phi(r_i) = -\frac{GM(r_i)}{r_i} - 4\pi G \sum_{r_j \leq r_i} \rho(r_j) r_j' \Delta r_j \quad (1.4)$$

$$\rho(r) = \rho_0 \exp \left(-2n \left[\left(\frac{r}{r_s} \right)^{\frac{1}{n}} - 1 \right] \right) \quad (1.5)$$

$$\beta_j(r) = 1 - \frac{\sigma_j^2(r)}{\sigma_r^2(r)} \quad (1.6)$$

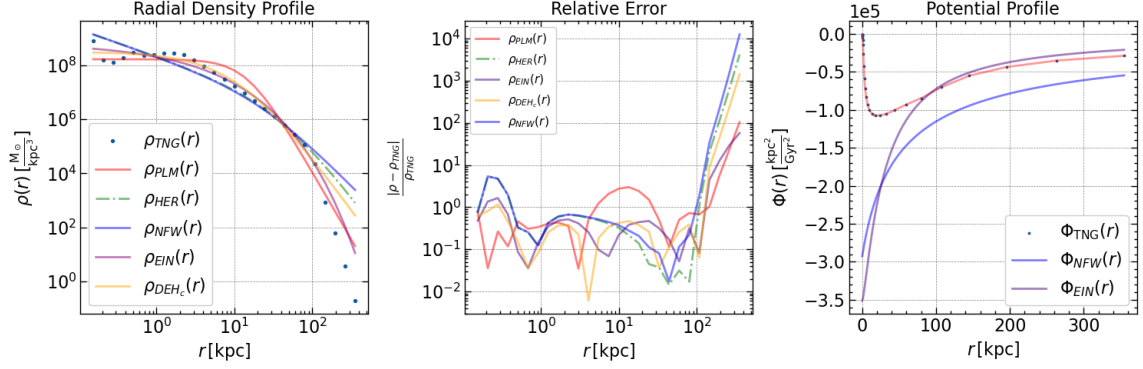


Figure 1.5: Radial density and potential profile of the dark matter distribution in the subhalo with corresponding fits using different analytical profiles. The figure in the middle showcases the relative error of the fit profile with respect to the data.

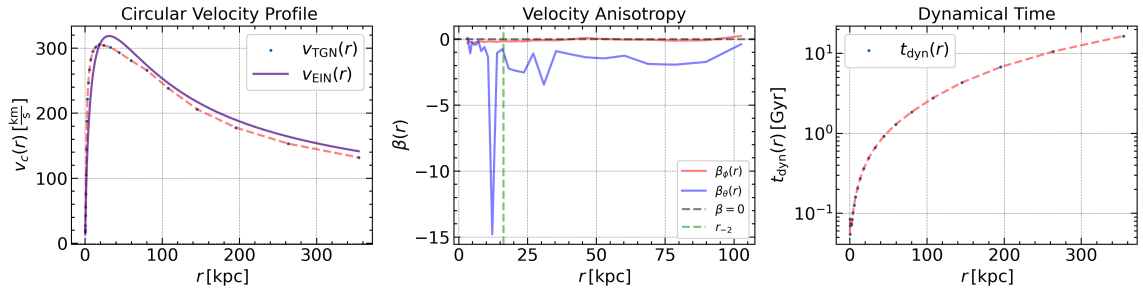


Figure 1.6: Profiles of circular velocity, velocity anisotropy, and dynamical time of the dark matter distribution in the subhalo.

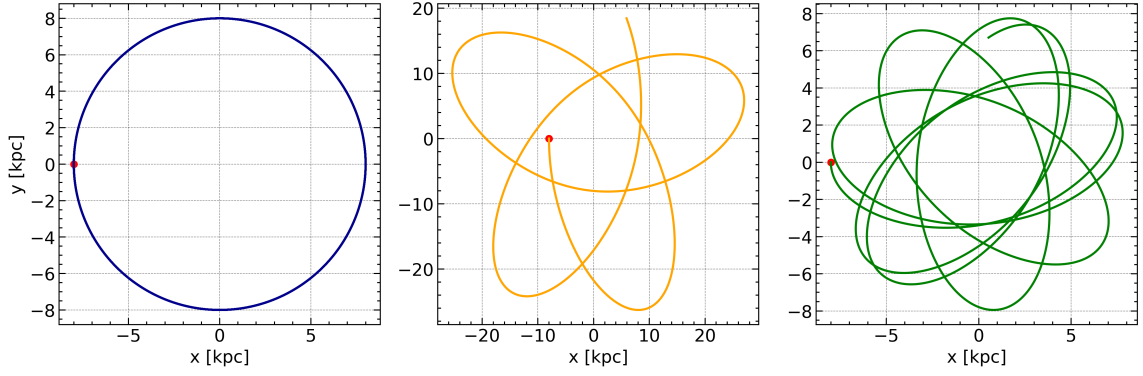


Figure 1.7: Orbits of a test particles initialized at a distance of $r_0 = 8$ kpc from the center of mass of the system with different initial circular velocity. The stable orbit (blue) corresponds to the circular velocity at this radius, while the other orbits (orange and green) show precessing rosette patterns due to their lower and higher initial velocity, respectively. The potential used for the orbit integration is the fitted Einasto potential of the system.

Appendix A

Appendix

Parameter	Value
Hubble parameter, h	0.6774
Matter density parameter, $\Omega_{m,0}$	0.3089
Dark energy density parameter, $\Omega_{\Lambda,0}$	0.6911
Baryonic matter density parameter, $\Omega_{b,0}$	0.0486
Simulation box size	$35000h^{-1}$ kpc
Number of dark matter particles	268525
Mass of individual dark matter particle	$10^{6.73}M_{\odot}$
Redshift of snapshot	$z \approx 0$

Table A.1: Cosmological parameters and constants relevant for the analysis of the subhalo..

Bibliography